

Engineering Mechanics: ME101

Lecture 4: Statics



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Contents

- ❖ Rigid Body Equilibrium,
- ❖ Free Body Diagram,
- ❖ Support Constraints and Reactions,
- ❖ Problems

Condition of Equilibrium for a Particle/Structure

- **Equilibrium**

- ✓ At rest remains at rest

- ✓ In motion with constant velocity

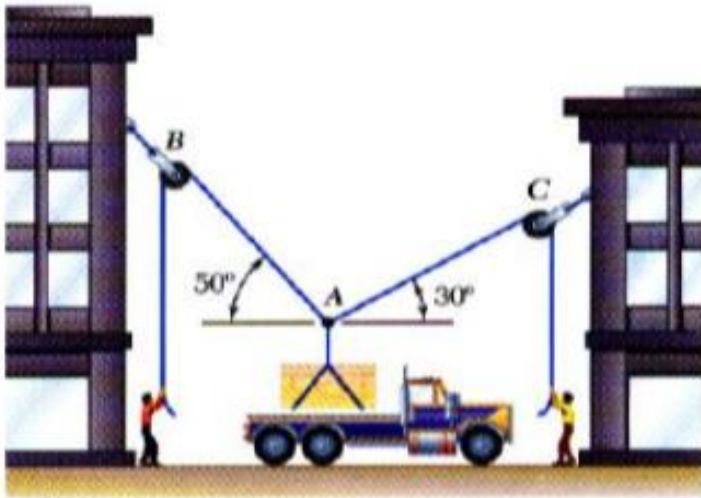
- **Static Equilibrium:** Most often an object at rest

- **In motion with constant velocity:** To maintain equilibrium, satisfy Newton's First Laws:

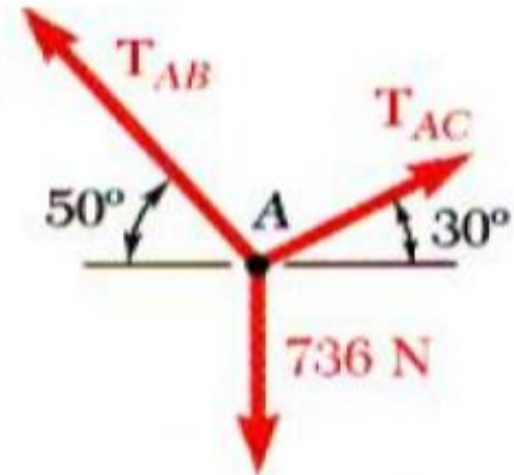
- ✓ Necessary and sufficient condition for equilibrium

- **First law:** The velocity of a body remains constant unless the body is acted upon by an external force.

Rigid Body Equilibrium & Free Body Diagram



Space Diagram: A sketch showing the physical conditions of the problem.



Free-Body Diagram: A sketch showing only the forces on the selected particle.

THE WHAT, WHY AND HOW OF A FREE BODY DIAGRAM (FBD)

What ? - It is a drawing that shows all external forces acting on the particle. It is the most important single step in the solution of problems in mechanics.

Why ? - It is key to being able to write the equations of equilibrium—which are used to solve for the unknowns (usually forces or angles).

□ Free Body Diagram (FBD):

- Must account for all known and unknown forces acting on the particle
- Isolate the body from its' surroundings
- Account for contact forces
- Account for forces like gravity; magnetic; electric forces.

Steps of Drawing a FBD

1. At the beginning, a clear decision is to be made by the analyst on the choice of the body to be considered for free body diagram.
2. Then that body is detached from all of its surrounding members including ground and only their forces on the free body are represented.
3. The weight of the body and other external body forces like centrifugal, inertia, etc., should also be included in the diagram and they are assumed to act at the centre of gravity of the body.
4. When a structure involving many elements is considered for free body diagram, the forces acting in between the elements should not be brought into the diagram.
5. The known forces acting on the body should be represented with proper magnitude and direction.
6. If the direction of unknown forces like reactions can be decided, they should be indicated clearly in the diagram.
7. Show coordinate directions on the diagram.

Two-dimensional reaction at supports and connections

❖ In general the support can be categories into 3 categories:

1. Hinged support

2. Free support or point or roller support

3. Fixed support

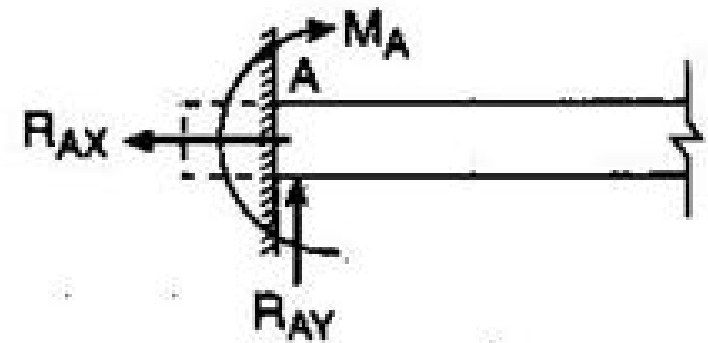
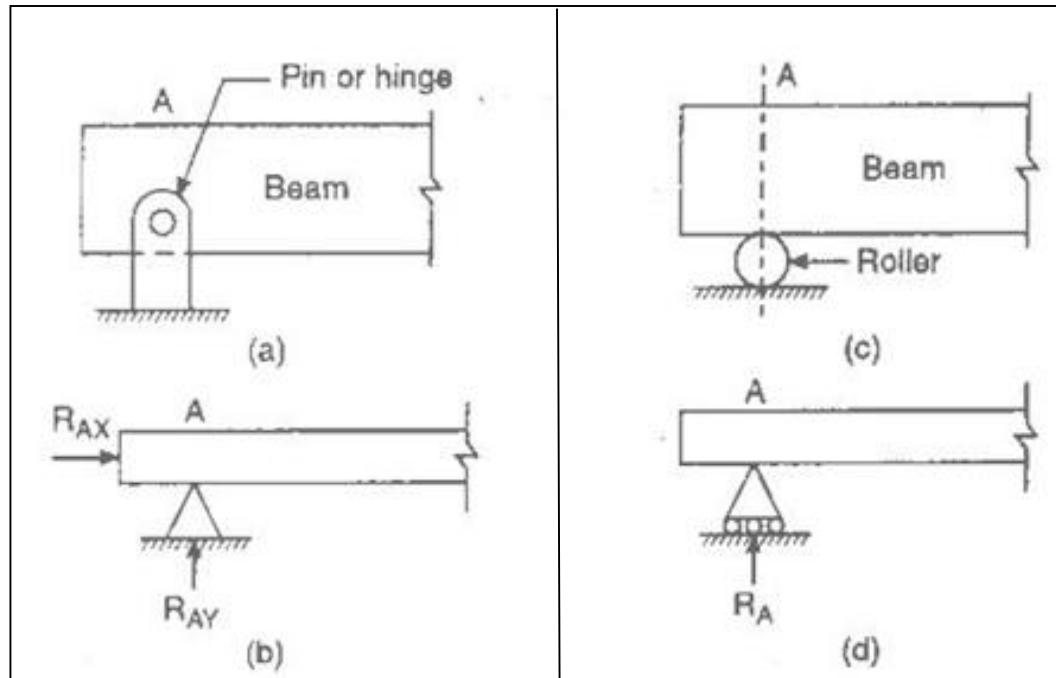
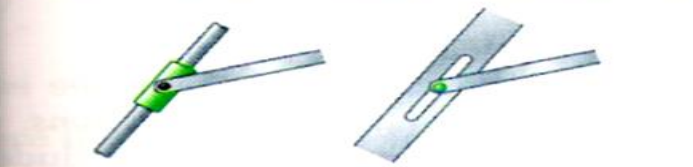
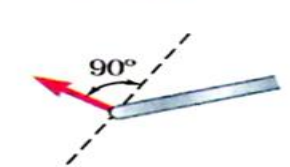
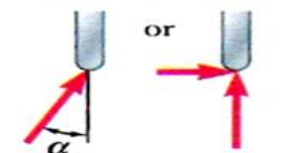
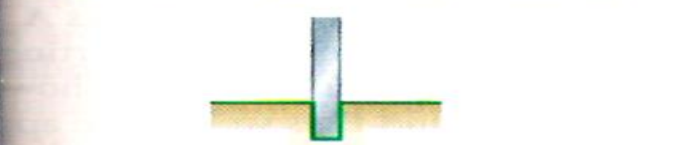
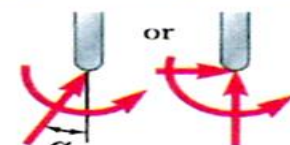


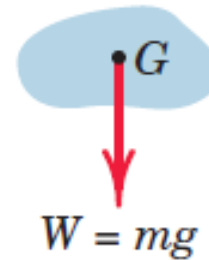
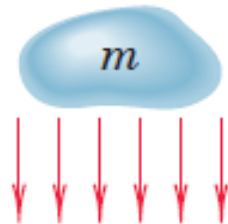
Fig. Fixed support (no translation and no rotation)

Fig. (a) & (b) Hinged support,
(c) & (d) Free or roller support

In brief: 2-D supports and connections with their reactions

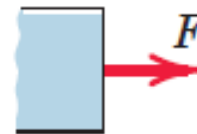
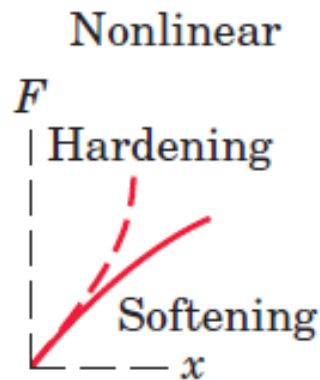
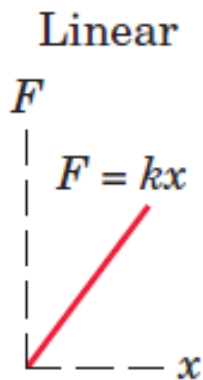
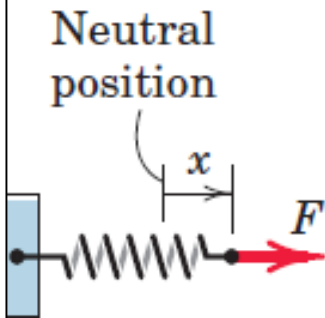
Support or Connection	Reaction	Number of Unknowns
 Rollers Rocker Frictionless surface	 Force with known line of action	1
 Short cable Short link	 Force with known line of action	1
 Collar on frictionless rod Frictionless pin in slot	 Force with known line of action	1
 Frictionless pin or hinge Rough surface	 Force of unknown direction	2
 Fixed support	 Force and couple	3

Gravitational attraction



The resultant of gravitational attraction on all elements of a body of mass m is the weight $W = mg$ and acts toward the center of the earth through the center mass G .

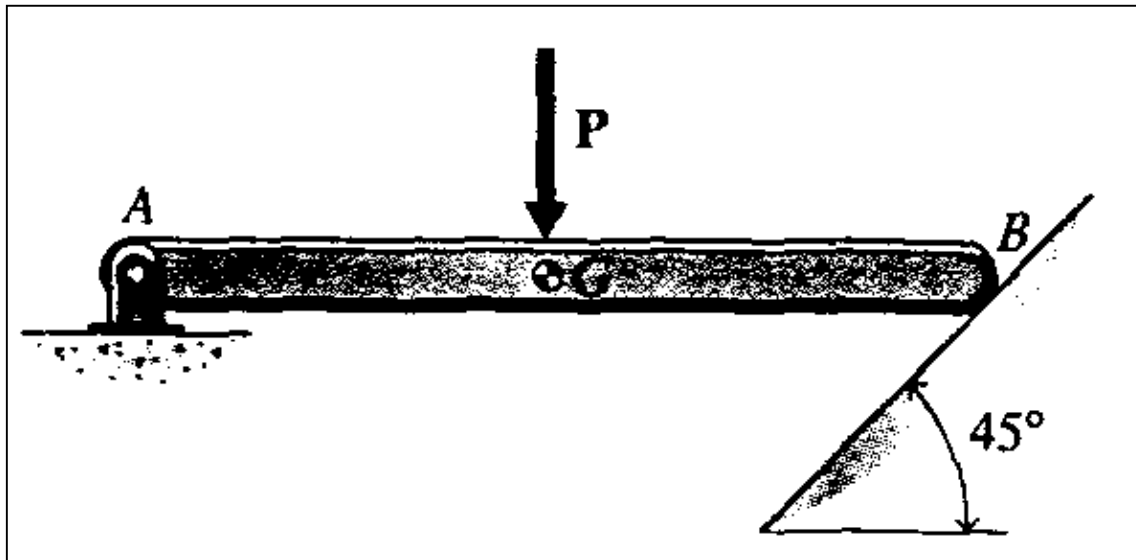
Spring action



Spring force is tensile if spring is stretched and compressive if compressed. For a linearly elastic spring the stiffness k is the force required to deform the spring a unit distance.

Example of FBD

Draw complete free-body diagram of the beam shown in Fig., which has a mass m .



Solution

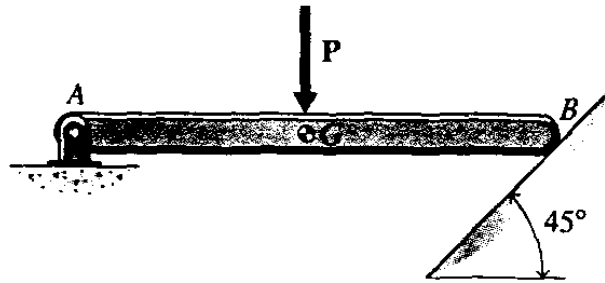
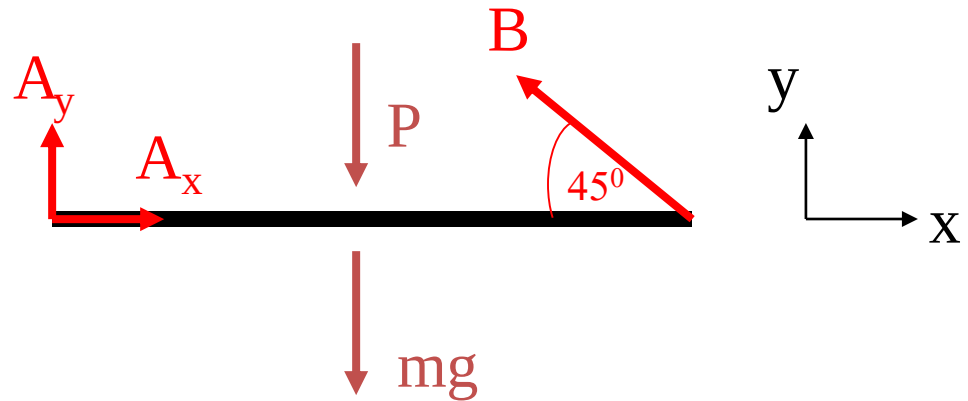


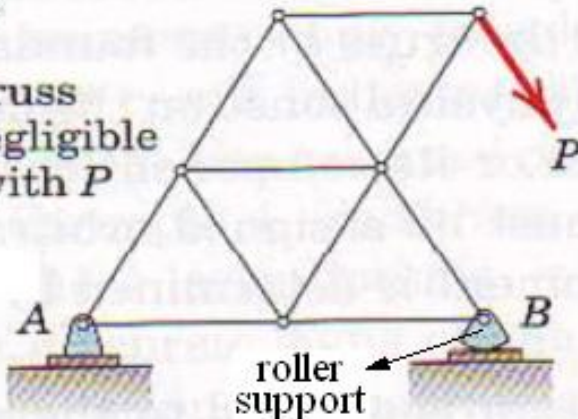
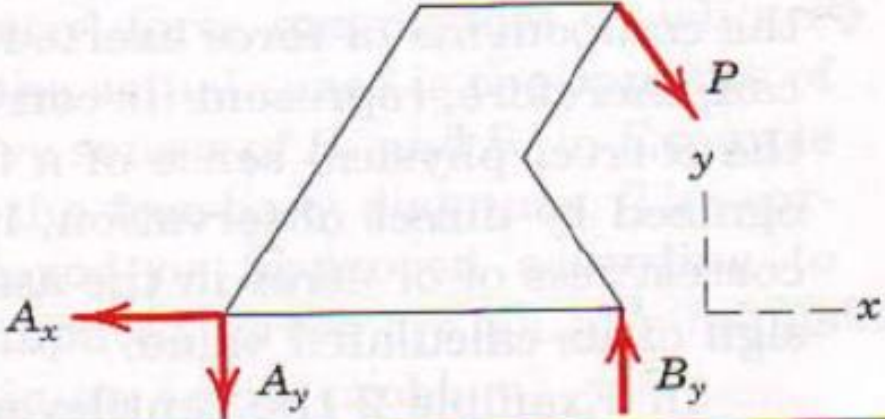
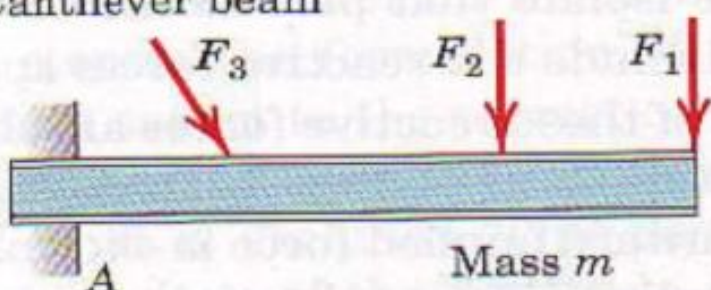
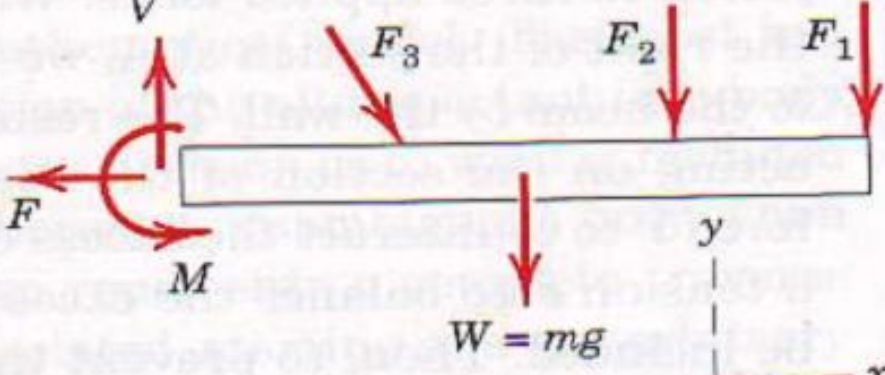
Fig. P6-2

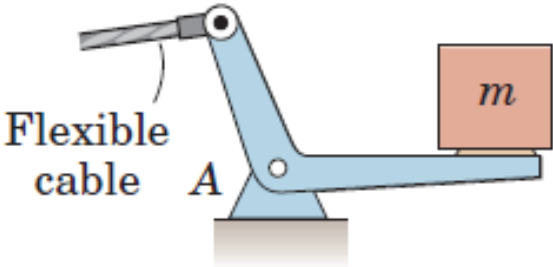
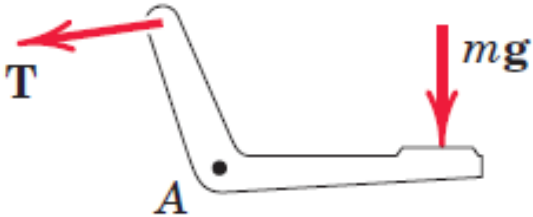
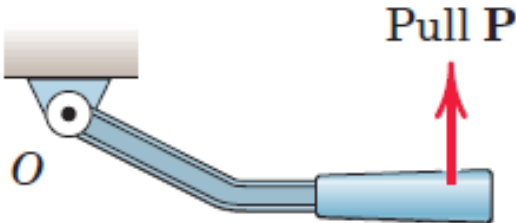
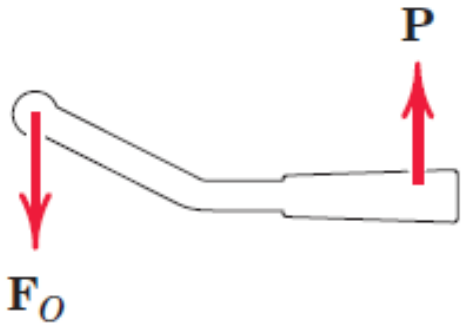
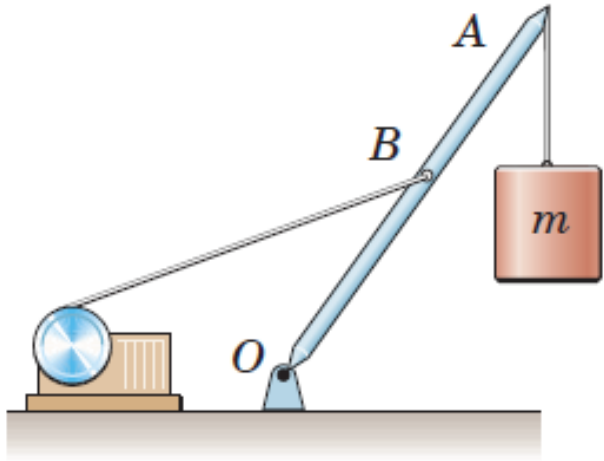
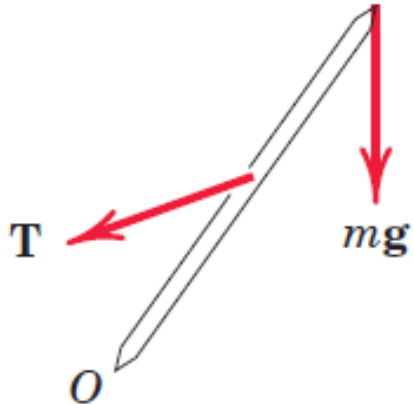
Known forces

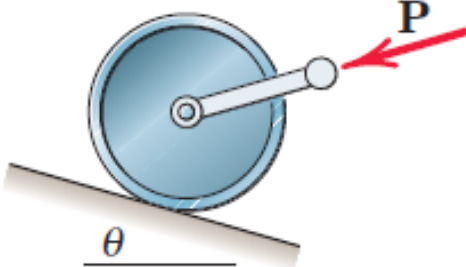
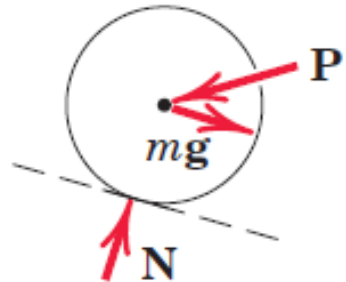
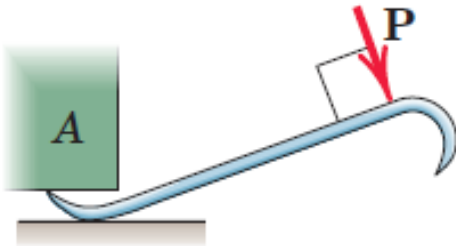
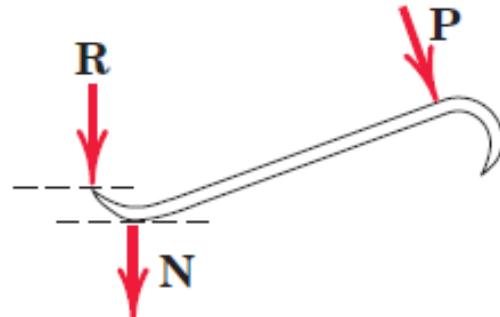
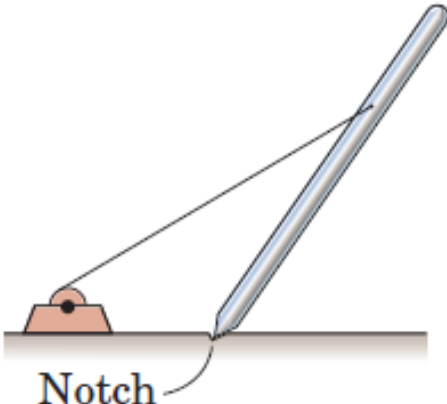
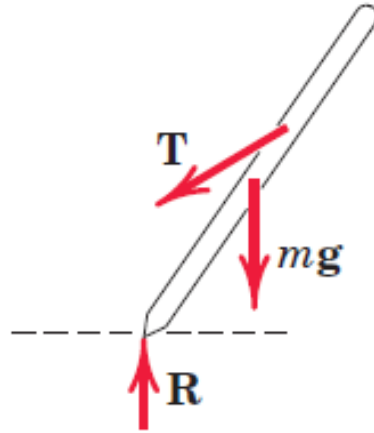
Unknown forces



Example of FBD (cont):

SAMPLE FREE-BODY DIAGRAMS	
Mechanical System	Free-Body Diagram of Isolated Body
1. Plane truss Weight of truss assumed negligible compared with P  A truss structure is shown with a pin support at point A and a roller support at point B. A downward force P is applied at the top right joint. The weight of the truss is assumed negligible compared with P.	 The free-body diagram shows the truss isolated. Reaction forces are A_x (horizontal, left), A_y (vertical, down) at point A, and B_y (vertical, up) at point B. The applied force P is shown acting downwards at the top right joint. A coordinate system with x and y axes is indicated.
2. Cantilever beam  A cantilever beam is shown fixed at point A. Three downward forces F_1, F_2, and F_3 are applied at different points along the beam. The mass of the beam is m.	 The free-body diagram shows the beam isolated. Reaction forces are V (vertical, up), M (moment, counter-clockwise), and F (horizontal, left) at point A. The applied forces F_1, F_2, and F_3 are shown acting downwards at their respective points. The weight $W = mg$ is shown acting downwards at the center of the beam. A coordinate system with x and y axes is indicated.

	Body	Incomplete FBD
1. Bell crank supporting mass m with pin support at A .		
2. Control lever applying torque to shaft at O .		
3. Boom OA , of negligible mass compared with mass m . Boom hinged at O and supported by hoisting cable at B .		

	Body	Wrong or Incomplete FBD
1. Lawn roller of mass m being pushed up incline θ .		
2. Prybar lifting body A having smooth horizontal surface. Bar rests on horizontal rough surface.		
3. Uniform pole of mass m being hoisted into position by winch. Horizontal supporting surface notched to prevent slipping of pole.		

Necessary Conditions of Equilibrium of a Rigid Body

The necessary condition for a rigid body to be in equilibrium is that the **resultant force** and **couple** acting on the rigid body must be zero.

$$\vec{R} = \sum_{i=1}^n \vec{F}_i = \vec{0}$$

$$\vec{M} = \sum_{i=1}^n \vec{M}_i = \vec{0}$$

Particle equilibrium

Rigid body equilibrium

These are **vector equations**.

3 independent equations in 2 D

$$R_x = \sum_{i=1}^n F_{xi} = 0$$

$$R_y = \sum_{i=1}^n F_{yi} = 0$$

$$M_z = \sum_{i=1}^n M_{zi} = 0$$

6 independent equations in 3D

$$R_x = \sum_{i=1}^n F_{xi} = 0$$

$$R_y = \sum_{i=1}^n F_{yi} = 0$$

$$R_z = \sum_{i=1}^n F_{zi} = 0$$

$$M_x = \sum_{i=1}^n M_{xi} = 0$$

$$M_y = \sum_{i=1}^n M_{yi} = 0$$

$$M_z = \sum_{i=1}^n M_{zi} = 0$$

Equilibrium of a Rigid Body in Three Dimensions

1. Six scalar equations are required to express the conditions for the equilibrium of a rigid body in the general three dimensional case.

$$\sum F_x = 0 \quad \sum F_y = 0 \quad \sum F_z = 0 \quad \text{(i)}$$

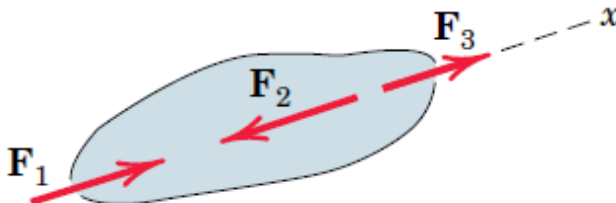
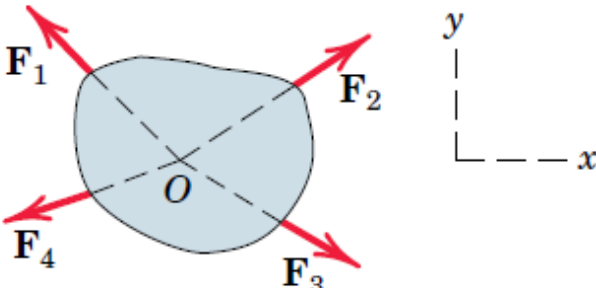
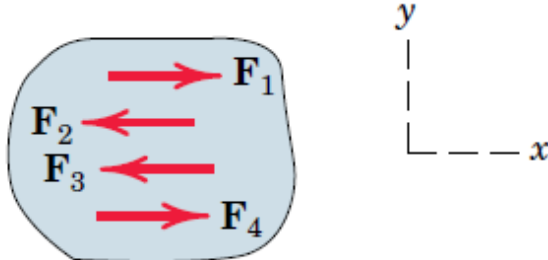
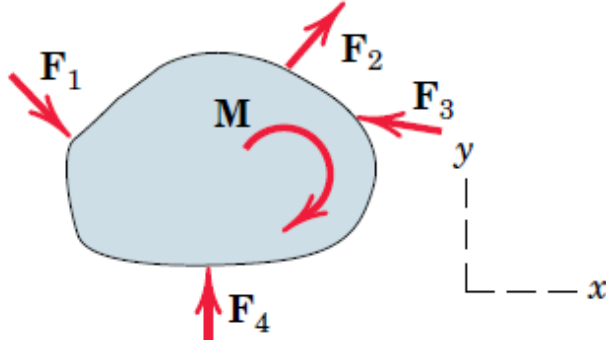
$$\sum M_x = 0 \quad \sum M_y = 0 \quad \sum M_z = 0 \quad \text{(ii)}$$

2. These equations can be solved for no more than 6 unknowns which generally represent reactions at supports or connections or unknown applied forces.

2. The scalar equations are conveniently obtained by applying the vector forms of the conditions for equilibrium,

$$\sum \mathbf{F} = 0 \quad \sum \mathbf{M}_O = \sum (\mathbf{r} \times \mathbf{F}) = 0 \quad \text{(iii)}$$

CATEGORIES OF EQUILIBRIUM IN TWO DIMENSIONS

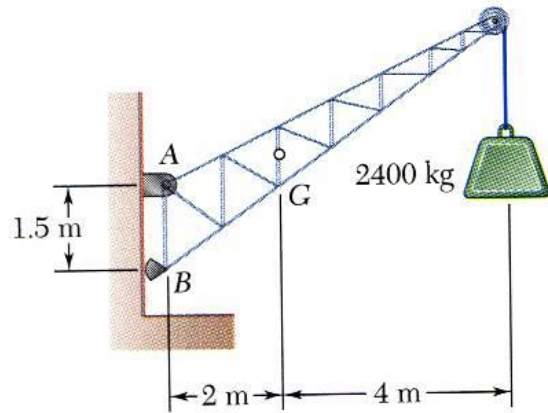
Force System	Free-Body Diagram	Independent Equations
1. Collinear		$\Sigma F_x = 0$
2. Concurrent at a point		$\Sigma F_x = 0$ $\Sigma F_y = 0$
3. Parallel		$\Sigma F_x = 0 \quad \Sigma M_z = 0$
4. General		$\Sigma F_x = 0 \quad \Sigma M_z = 0$ $\Sigma F_y = 0$

How to Solve Equilibrium Problems

1. Draw a proper free-body diagram.
2. Resolve force vectors into x- and y-components, if necessary.
3. Write an expression for the vector sum of the left-right forces. Then write an expression for the vector sum of the up-down forces. Set each of these expressions equal to zero.
4. If the no. of unknown more then sometimes we have to use equation of couple.
5. Solve the resulting algebraic equations.

Application of FBD and Equation of Equilibrium (EOE)

Sample Problem 1:

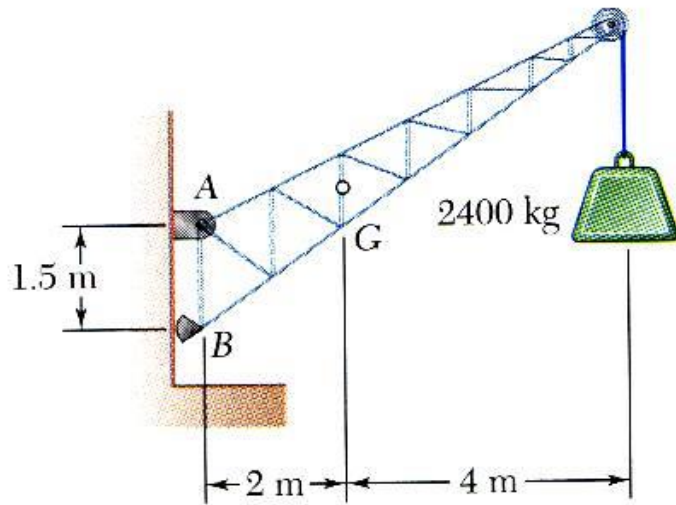


A fixed crane has a mass of 1000 kg and is used to lift a 2400 kg crate. It is held in place by a pin at A and a rocker at B. The center of gravity of the crane is located at G.

Determine the components of the reactions at A and B.

1. Determine B by solving the equation for the sum of the moments of all forces about A.
2. Determine the reactions at A by solving the equations for the sum of all horizontal force components and all vertical force components.
2. Check the values obtained for the reactions by verifying that the sum of the moments about B of all forces is zero.

Example 1 of FBD and EOE (*cont*):



$$R_x = \sum_{i=1}^n F_{xi} = 0$$

$$R_y = \sum_{i=1}^n F_{yi} = 0$$

Figure (1)

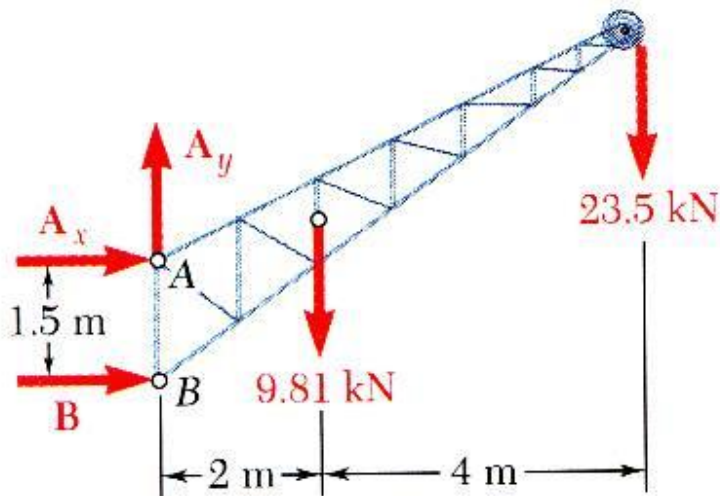


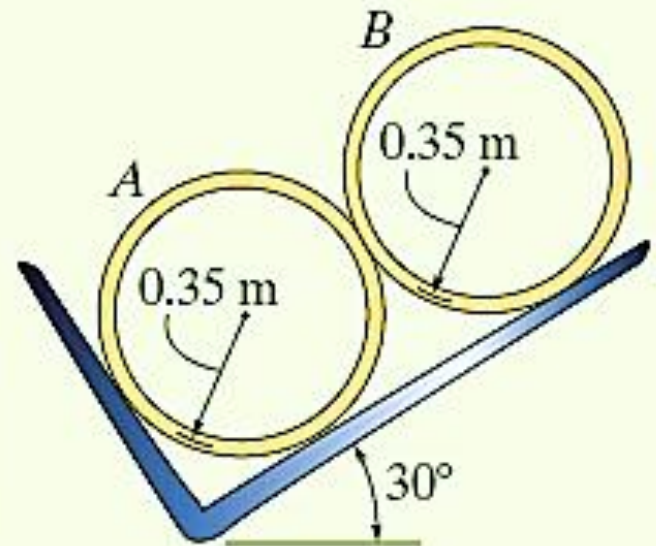
Figure (2)

$$M_z = \sum_{i=1}^n M_{zi} = 0$$

Free body diagram



(a)



(b)

Free body diagram example

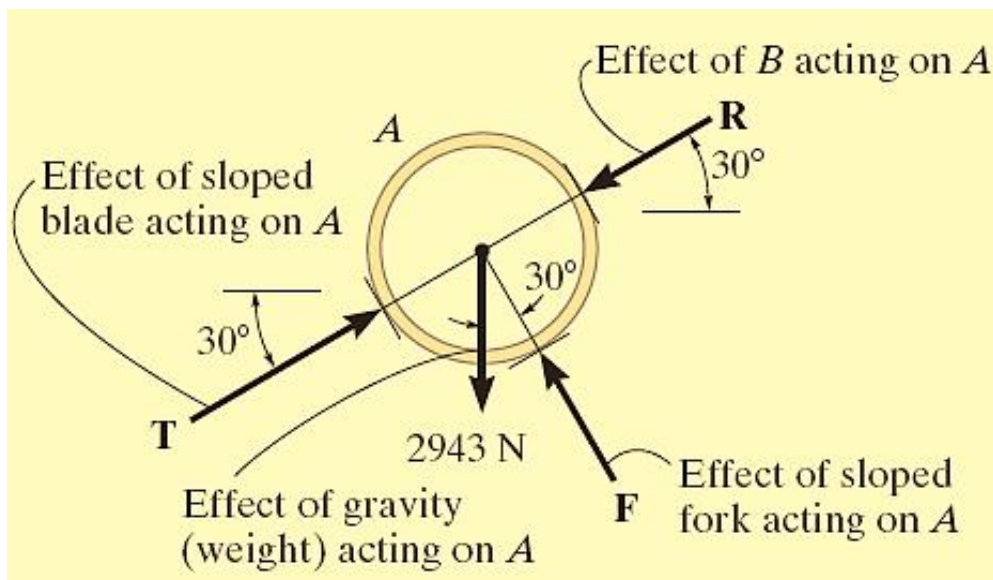
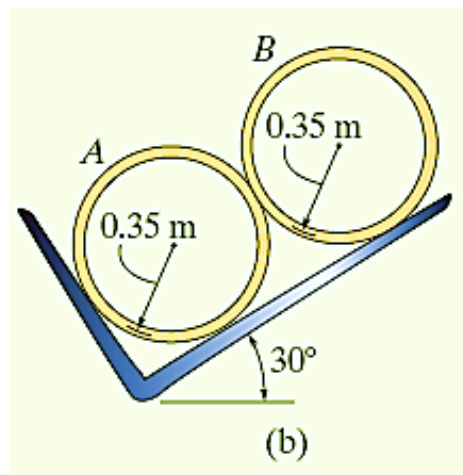


Figure (1)

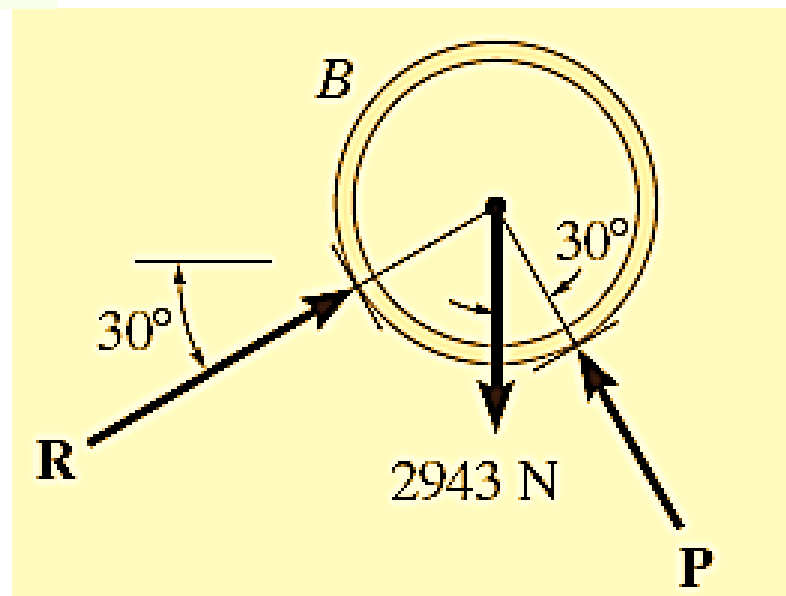
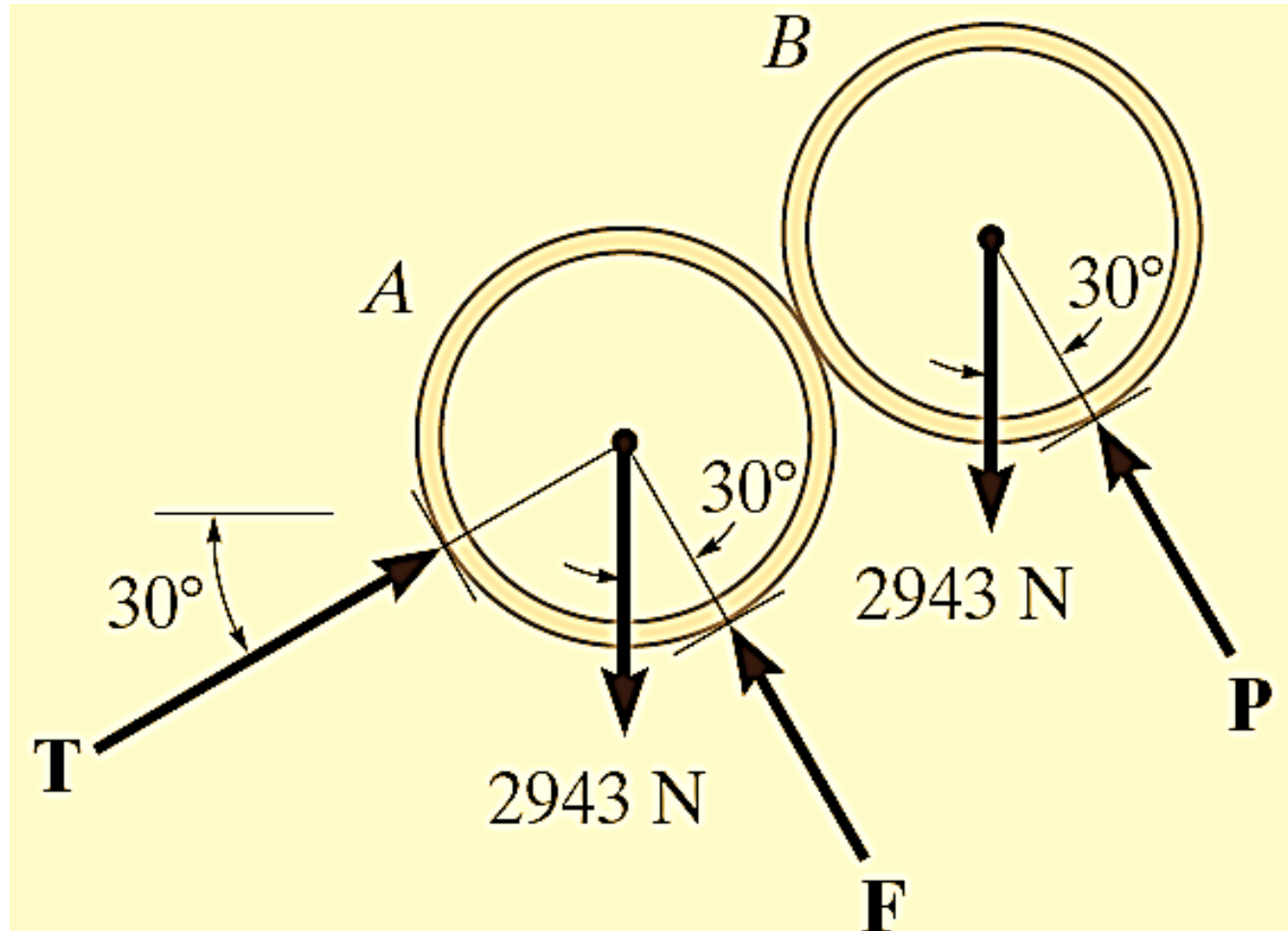
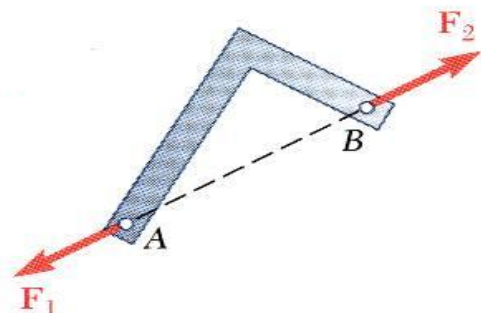
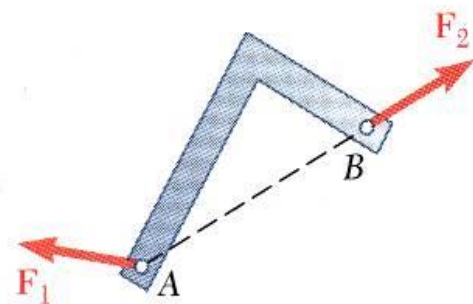
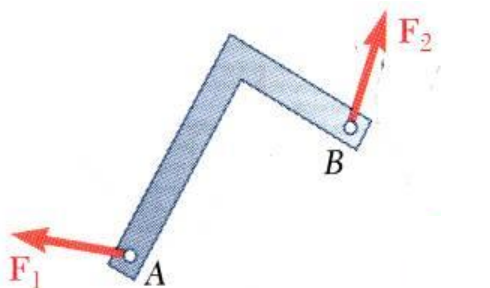


Figure (2)

Free body diagram example

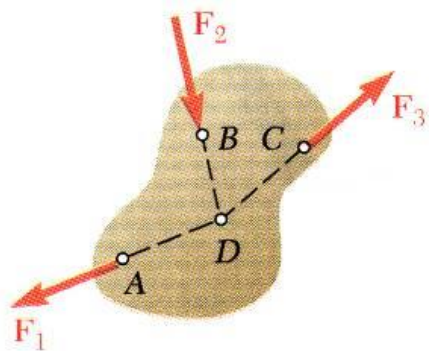
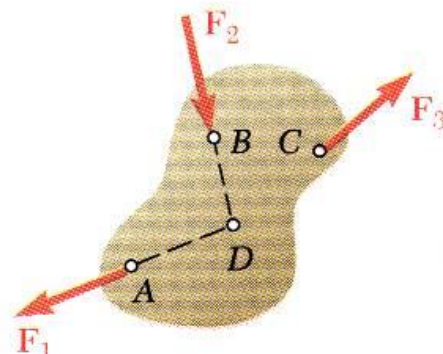
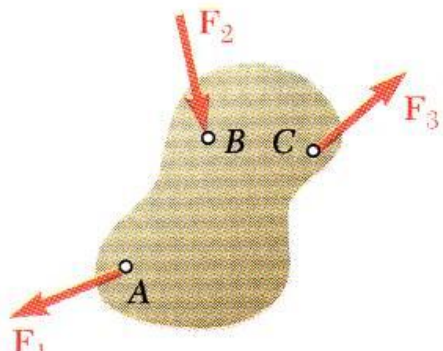


Equilibrium of a Two-Force Body



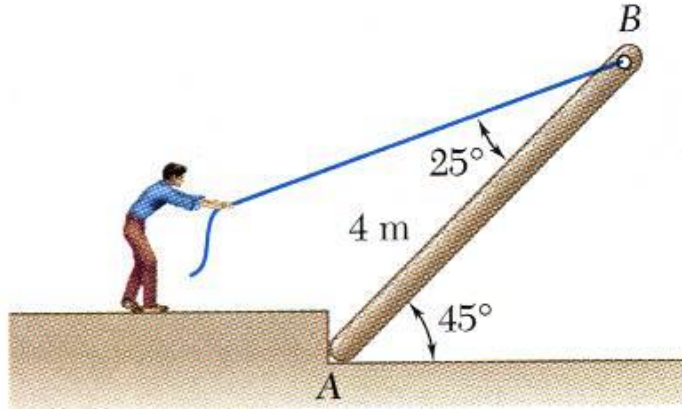
- Consider a plate subjected to two forces F_1 and F_2 .
1. For static equilibrium, the sum of moments about A must be zero. The moment of F_2 must be zero. It follows that the line of action of F_2 must pass through A.
 2. Similarly, the line of action of F_1 must pass through B for the sum of moments about B to be zero.
 3. Requiring that the sum of forces in any direction be zero leads to the conclusion that F_1 and F_2 must have equal magnitude but opposite sense.

Equilibrium of a Three-Force Body



- Consider a rigid body subjected to forces acting at only 3 points.
1. Assuming that their lines of action intersect, the moment of $\mathbf{F}_1$ and $\mathbf{F}_2$ about the point of intersection represented by D is zero.
 2. Since the rigid body is in equilibrium, the sum of the moments of $\mathbf{F}_1$, $\mathbf{F}_2$, and $\mathbf{F}_3$ about any axis must be zero. It follows that the moment of $\mathbf{F}_3$ about D must be zero as well and that the line of action of $\mathbf{F}_3$ must pass through D .
 3. The lines of action of the three forces must be concurrent or parallel.

Sample Problem



- ❖ A man raises a 10-kg joist, of length 4 m, by pulling on a rope.
- ✓ Find the tension T in the rope and the reaction at A.

SOLUTION:

- Create a free-body diagram of the joist. Note that the joist is a 3 force body acted upon by the rope, its weight, and the reaction at A.
1. The three forces must be concurrent for static equilibrium. Therefore, the reaction $\mathbf{R}$ must pass through the intersection of the lines of action of the weight and rope forces. Determine the direction of the reaction $\mathbf{R}$.
 2. Utilize a force triangle to determine the magnitude of the reaction $\mathbf{R}$.

Sample Problem

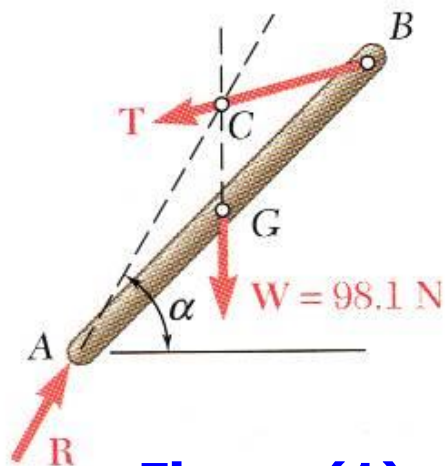


Figure (1)

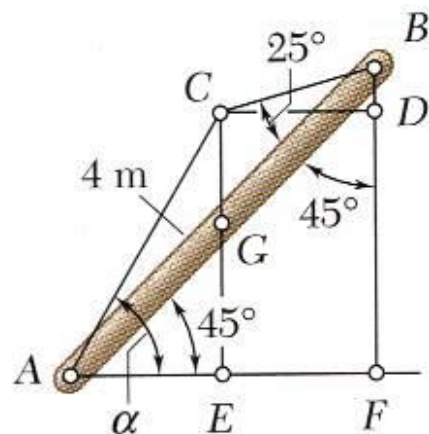


Figure (2)

- Create a free-body diagram of the joist.

- Determine the direction of the reaction **R**.

$$BF = AB \cos 45^\circ = (4 \text{ m}) \cos 45^\circ = 2.828 \text{ m}$$

$$CD = AE = \frac{1}{2} AF = 1.414 \text{ m}$$

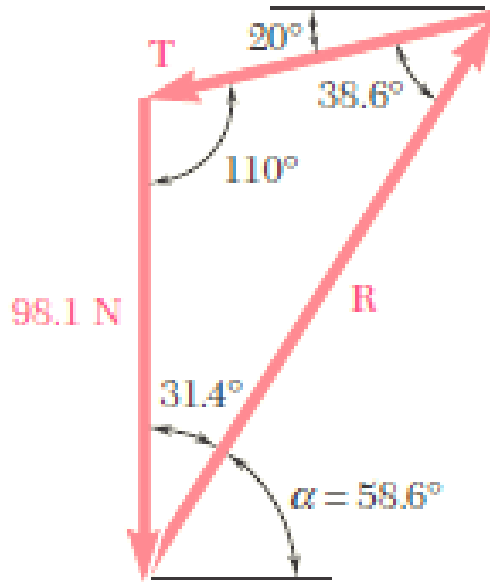
$$BD = CD \cot(45^\circ + 25^\circ) = (1.414 \text{ m}) \tan 20^\circ = 0.515 \text{ m}$$

$$CE = BF - BD = (2.828 - 0.515) \text{ m} = 2.313 \text{ m}$$

$$\tan \alpha = \frac{CE}{AE} = \frac{2.313}{1.414} = 1.636$$

$$\alpha = 58.6^\circ$$

Sample Problem



- Determine the magnitude of the reaction R .

$$\frac{T}{\sin 31.4^\circ} = \frac{R}{\sin 110^\circ} = \frac{98.1 \text{ N}}{\sin 38.6^\circ}$$

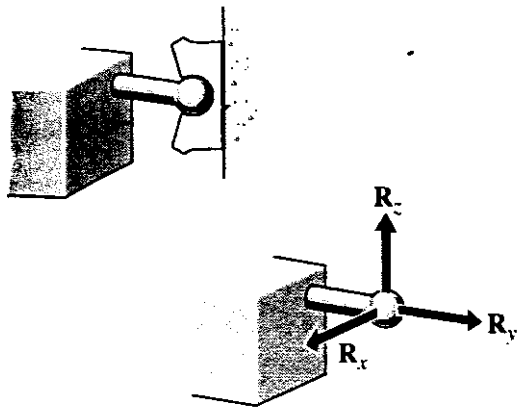
$$T = 81.9 \text{ N}$$

$$R = 147.8 \text{ N}$$

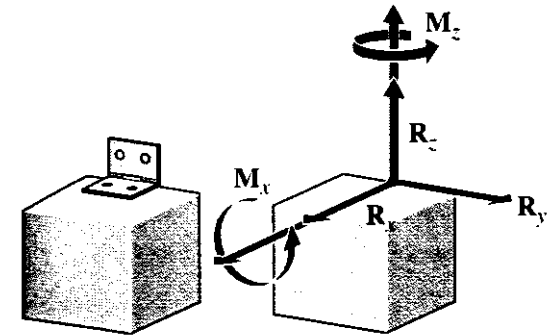
3-D Case

Three-dimensional reaction at supports and connections

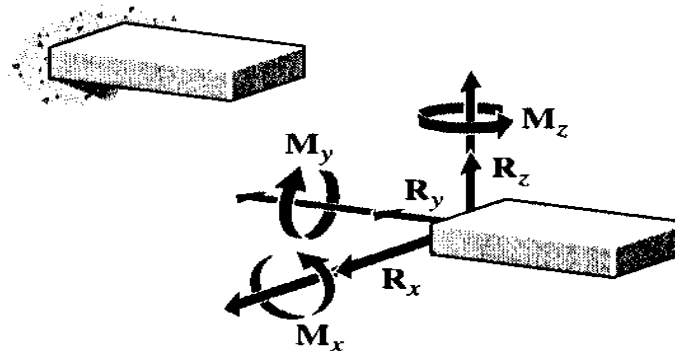
1. Ball and socket



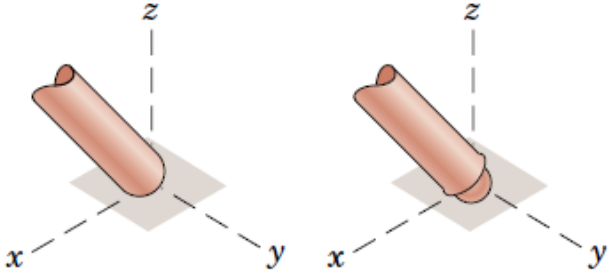
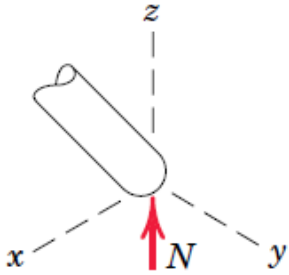
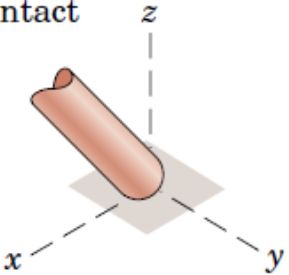
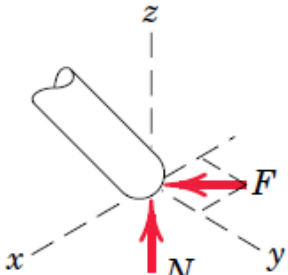
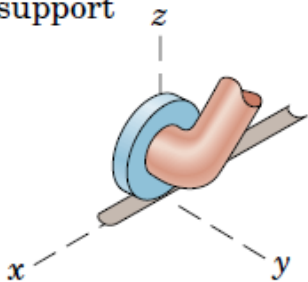
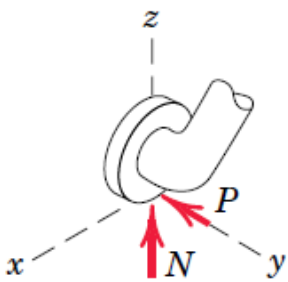
2. Hinge



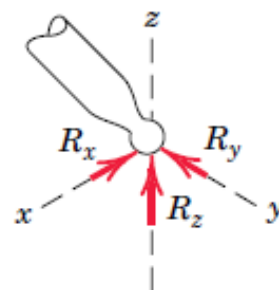
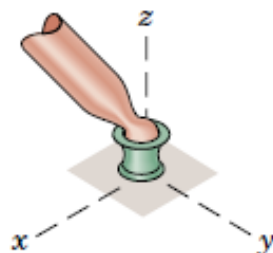
3. Fixed



MODELING THE ACTION OF FORCES IN THREE-DIMENSIONAL ANALYSIS

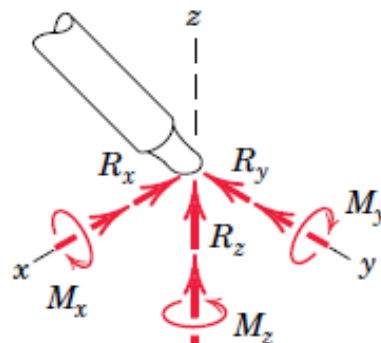
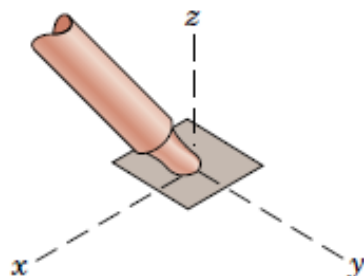
Type of Contact and Force Origin	Action on Body to Be Isolated
1. Member in contact with smooth surface, or ball-supported member 	 Force must be normal to the surface and directed toward the member.
2. Member in contact with rough surface 	 The possibility exists for a force F tangent to the surface (friction force) to act on the member, as well as a normal force N.
3. Roller or wheel support with lateral constraint 	 A lateral force P exerted by the guide on the wheel can exist, in addition to the normal force N.

4. Ball-and-socket joint



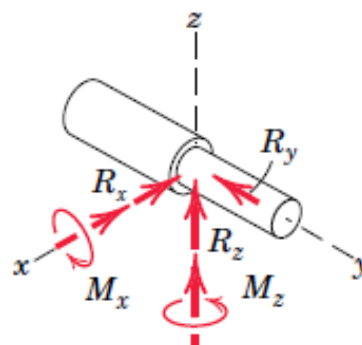
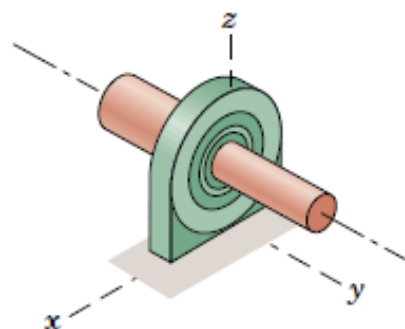
A ball-and-socket joint free to pivot about the center of the ball can support a force $\mathbf{R}$ with all three components.

5. Fixed connection (embedded or welded)

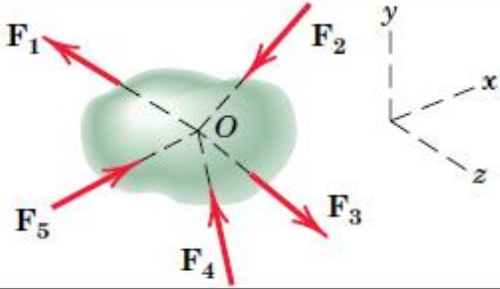
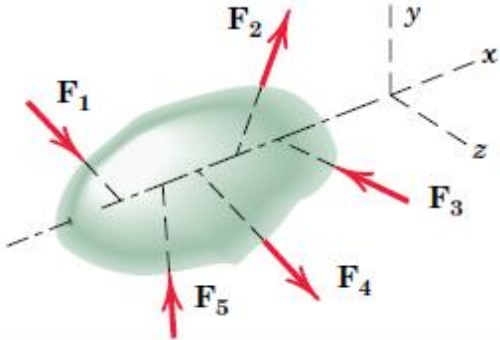
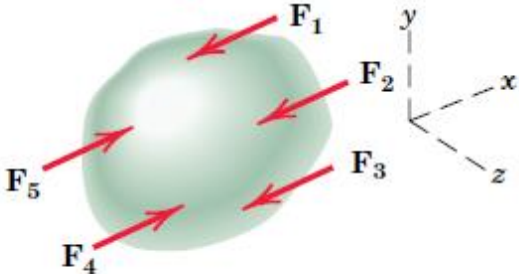
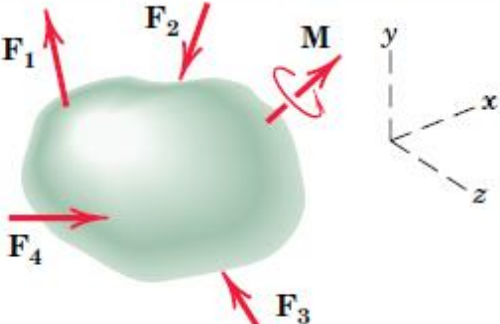


In addition to three components of force, a fixed connection can support a couple $\mathbf{M}$ represented by its three components.

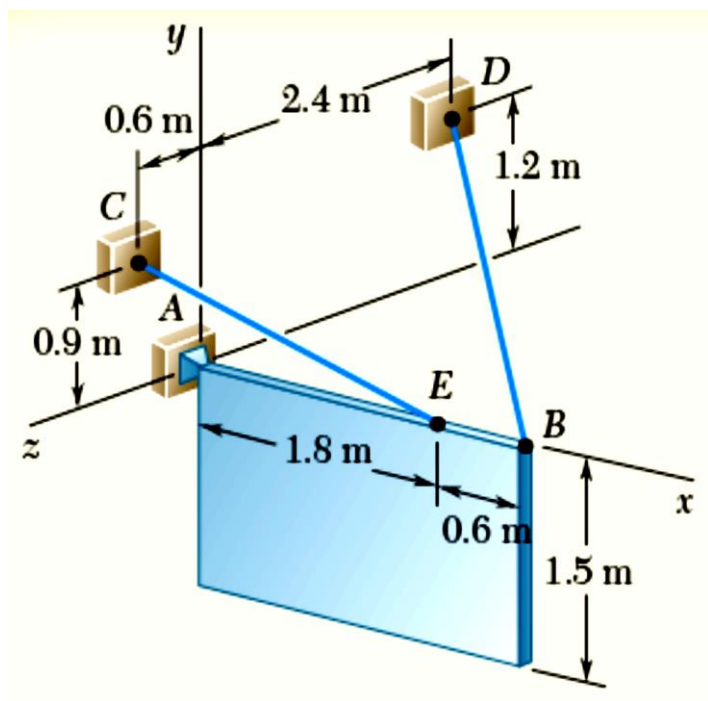
6. Thrust-bearing support



Thrust bearing is capable of supporting axial force R_y as well as radial forces R_x and R_z . Couples M_x and M_z must, in some cases, be assumed zero in order to provide statical determinacy.

CATEGORIES OF EQUILIBRIUM IN THREE DIMENSIONS		
Force System	Free-Body Diagram	Independent Equations
1. Concurrent at a point		$\Sigma F_x = 0$ $\Sigma F_y = 0$ $\Sigma F_z = 0$
2. Concurrent with a line		$\Sigma F_x = 0$ $\Sigma M_y = 0$ $\Sigma F_y = 0$ $\Sigma M_z = 0$ $\Sigma F_z = 0$
3. Parallel		$\Sigma F_x = 0$ $\Sigma M_y = 0$ $\Sigma M_z = 0$
4. General		$\Sigma F_x = 0$ $\Sigma M_x = 0$ $\Sigma F_y = 0$ $\Sigma M_y = 0$ $\Sigma F_z = 0$ $\Sigma M_z = 0$

Sample Problem



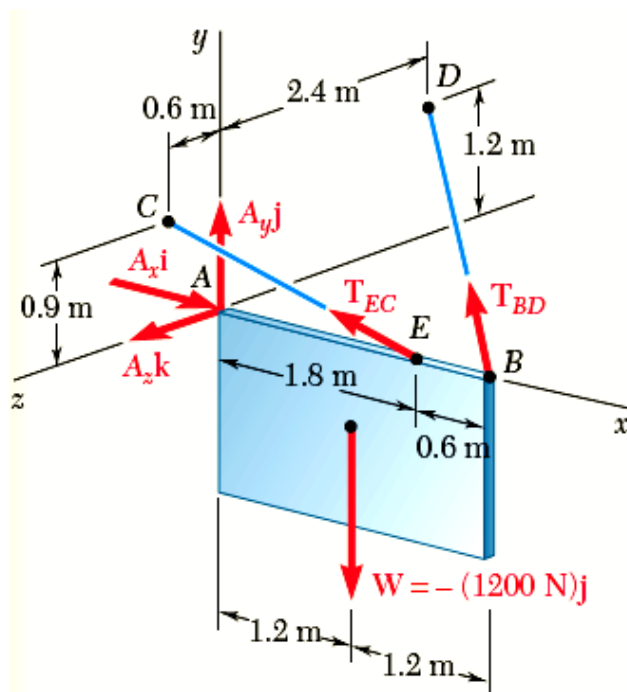
SOLUTION:

- Create a free-body diagram for the sign.
- Apply the conditions for static equilibrium to develop equations for the unknown reactions.

❑ A sign of uniform density weighs 1200 N and is supported by a ball-and-socket joint at A and by two cables.

✓ Determine the tension in each cable and the reaction at A.

Sample Problem



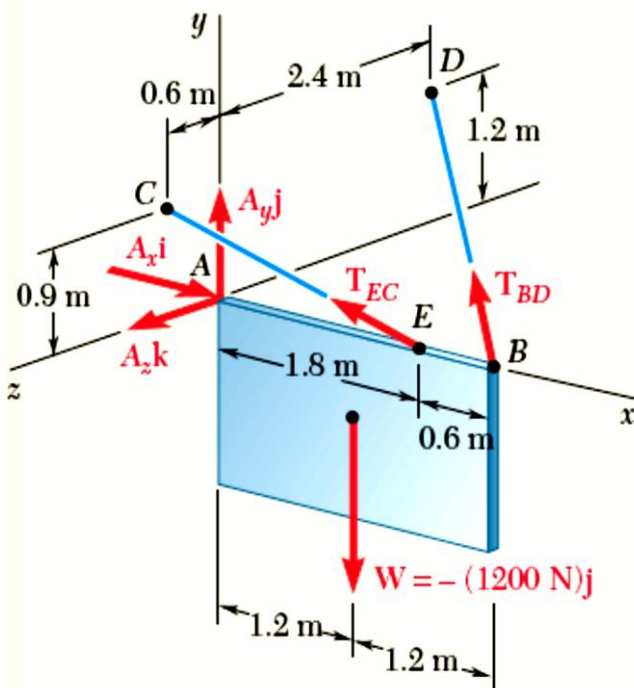
- Create a free-body diagram for the sign.

Since there **are only 5 unknowns**, the sign is partially constrained. **It is free to rotate about the x axis.** It is, however, in equilibrium for the given loading.

$$\begin{aligned}\mathbf{T}_{BD} &= T_{BD} \frac{\overrightarrow{BD}}{BD} & \text{(i)} \\ &= T_{BD} \left(-\frac{2}{3}\mathbf{i} + \frac{1}{3}\mathbf{j} - \frac{2}{3}\mathbf{k} \right)\end{aligned}$$

$$\begin{aligned}\mathbf{T}_{EC} &= T_{EC} \frac{\overrightarrow{EC}}{EC} & \text{(ii)} \\ &= T_{EC} \left(-\frac{6}{7}\mathbf{i} + \frac{3}{7}\mathbf{j} + \frac{2}{7}\mathbf{k} \right)\end{aligned}$$

Sample Problem



$$\sum \mathbf{F} = \mathbf{A} + \mathbf{T}_{BD} + \mathbf{T}_{EC} - (1200 \text{ N})\mathbf{j} = 0 \quad (\text{iii})$$

$$\mathbf{i}: A_x - \frac{2}{3}T_{BD} - \frac{6}{7}T_{EC} = 0$$

$$\mathbf{j}: A_y + \frac{1}{3}T_{BD} + \frac{3}{7}T_{EC} - 1200 \text{ N} = 0$$

$$\mathbf{k}: A_z - \frac{2}{3}T_{BD} + \frac{2}{7}T_{EC} = 0$$

$$\sum \mathbf{M}_A = \mathbf{r}_B \times \mathbf{T}_{BD} + \mathbf{r}_E \times \mathbf{T}_{EC} + (1.2 \text{ m})\mathbf{i} \times (-1200 \text{ N})\mathbf{j} = 0 \quad (\text{iv})$$

$$\mathbf{j}: 1.6T_{BD} - 0.514T_{EC} = 0$$

$$\mathbf{k}: 0.8T_{BD} + 0.771T_{EC} - 1440 \text{ N} \cdot \text{m} = 0$$

- Apply the conditions for static equilibrium to develop equations for the unknown reactions.

Solve the 5 equations for the 5 unknowns,

$$T_{BD} = 450 \text{ N} \quad T_{EC} = 1400.8 \text{ N}$$

$$\mathbf{A} = (1500.7 \text{ N})\mathbf{i} + (449.7 \text{ N})\mathbf{j} - (100.2 \text{ N})\mathbf{k}$$

End